

EpiParam

User's Guide

English Version

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1. Introduction

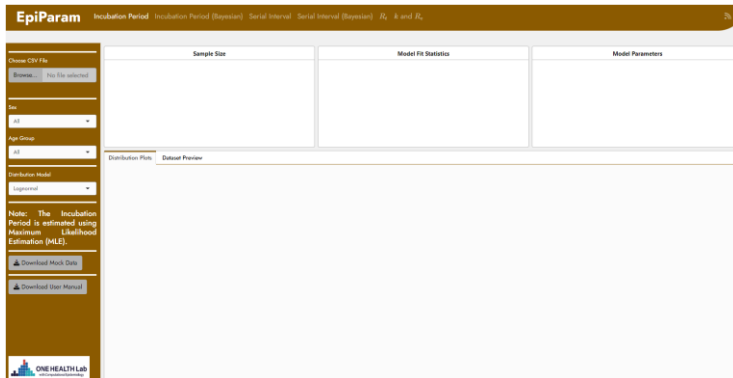
EpiParam (Epidemiological Parameter Estimation for Infectious Diseases) is a Shiny interface for estimating five main parameters (incubation period, serial interval, R_0 , R_t , and k) in the epidemiology of infectious diseases, developed by the OneHealth laboratory of the Department of Preventive Medicine at The Catholic University of Korea, Seoul, Korea. EpiParam introduces the Bayesian parametric model component that estimates the incubation period and serial interval when the exposure date and dates of onset are intervals.

This user's guide provides instructions on how to use the EpiParam application to generate the incubation period and serial interval, using the conventional parametric models (lognormal, Weibull, gamma and normal) and the Bayesian parametric models. It also demonstrates the estimation of the reproduction number (R_0), and the Kappa value (k) for a given infectious disease.

EpiParam is published in the GitHub repository <https://github.com/ACHANGWA/Epiparam>

To begin, open the GitHub repository and download all the supporting files into a folder on your desktop. Open the RMD file ([Epiparam_Full_App_w_Bayesian_02-02-2026.Rmd](#)) and run it in Rstudio in your computer.

Note that for a smooth functioning of the application, all the downloaded files from the repository must be place in a single folder in your computer.



This application was built to estimate five parameters: The incubation period, serial interval, reproduction number (R_0), and the Kappa value (k). It estimates the incubation period and serial interval using both the conventional parametric models and the Bayesian models.

2.0 Incubation period

This is the period between exposure/infection of an individual by a pathogen and the manifestation of the signs and symptoms of the disease. To estimate the incubation period, click on the ‘Incubation period’ window in the application.

Data preparation.

1. Save your data spreadsheet in a ‘.csv’ format example: “Busan_data.csv”
2. In the CSV data spreadsheet, the following variables should be present with the given specificities (Capital/small letter sensitive);

- ✓ *Date of exposure of the disease → rename as “date_of_exposure”
- ✓ *Date of symptom onset → rename as “date_of_onset”
- ✓ Gender/Sex → rename as “sex”
- ✓ Age/Age-group → rename as “age_group”
- ✓ Any other variable depending on the researchers’ interests.

Mock data for the COVID-19 outbreak.

Russian_real.csv - LibreOffice Calc

A	B	C	D	E	F	G	H	I	J	K
1	id	date_of_exposure	regions	date_of_onset	Source of infection	type	country	Transmission type	Age group	sex
2	#1	2021-11-24	incheon	2021-11-25	11.24 Arrivals from Nigeria(E1672)	abroad	Nigeria	Imported	41M	
3	#2	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	45F	
4	#3	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	43M	
5	#4	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	25F	
6	#5	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	51M	
7	#6	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	43F	
8	#7	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	61M	
9	#8	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	33M	
10	#9	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	42F	
11	#10	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40F	
12	#11	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	34F	
13	#12	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	55F	
14	#13	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	44M	
15	#14	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	46M	
16	#15	2021-11-25	Daejeon	2021-11-26	Patient #1	National	Domestic	Domestic	31F	
17	#16	2021-11-26	Daejeon	2021-11-26	Patient #15	National	Domestic	Domestic	47M	
18	#17	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	39F	
19	#18	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	48M	
20	#19	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	51F	
21	#20	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	60M	
22	#21	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	53M	
23	#22	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	46F	
24	#23	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	56F	
25	#24	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	44F	
26	#25	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	23F	
27	#26	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	25M	
28	#27	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	47M	
29	#28	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	49F	
30	#29	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	37M	
31	#30	2021-11-26	Daejeon	2021-11-29	Patient #29	National	Domestic	Domestic	42F	
32	#31	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	43M	
33	#32	2021-11-26	Daejeon	2021-11-30	Patient #1	National	Domestic	Domestic	59F	
34	#33	2021-11-26	Daejeon	2021-11-30	Patient #1	National	Domestic	Domestic	51M	

This dataset can be downloaded from the ‘Download Mock data’ icon on the incubation period page after running the RMD.

Required formats

- ❖ date_of_exposure

This is the probable date of exposure of the case, in the format “YYYY-MM-DD”

- ❖ date_of_onset

This is the probable date of symptom onset of the disease, in the format “YYYY-MM-DD”

❖ sex

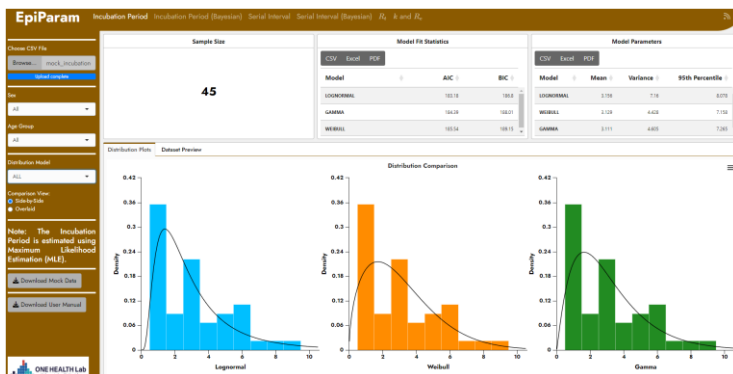
the gender of the individuals; All = Everyone, regardless of gender;
Male = M; Female = F

❖ age

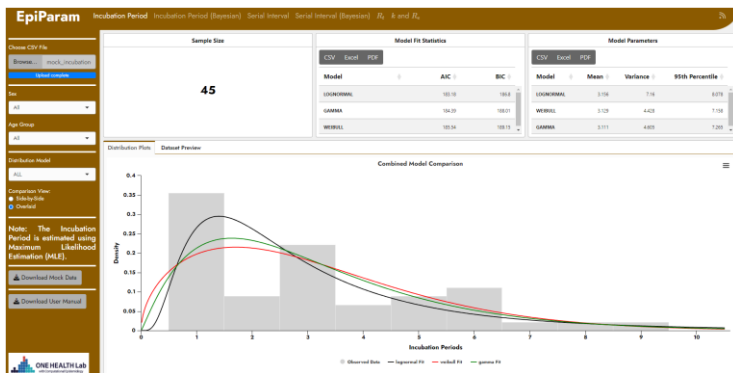
The ages of the participants should be numeric in years.
Classifications of the age into age include: All is everyone
regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

A) Estimation using conventional parametric models

To estimate the incubation period.



1. Click on the 'Incubation Period' icon at the top of the App.
2. Upload the .csv file containing the cleaned data you wish to use, by clicking on "Browse" and selecting the data from the computer.
3. Select which parametric model you wish to use (lognormal, Weibull, gamma or ALL)
4. In case you select 'ALL', select the comparison view. The view can either be 'side by side' or 'overlaid'.



Output Description

The output window is made up of 5 sub-windows.

- The 'Sample Size' window gives the total number of rows (individuals/patients/units) in the uploaded dataset.
- The 'Dataset Preview' window previews the uploaded dataset.
- The 'Model Fit Statistics' window produces the model AIC and BIC.
- The 'Model Parameters' window, provides estimates (mean, variance, 95th percentile using the respective models.)
- The 'Distribution plots' window, shows the output histogram of the distributions.

Sub-group analysis

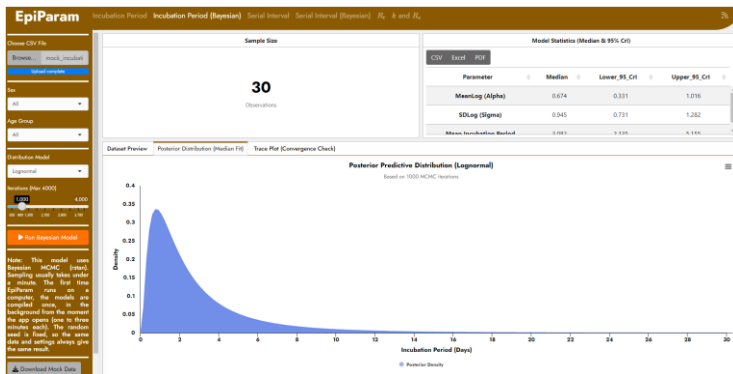
Based on individual needs, select stratifications by sex or by age group or a combination of both.

B) Estimation using the Bayesian parametric models

The Bayesian parametric models are used to estimate the incubation period when there exists an uncertainty in the exposure dates, resulting in the dates being an interval.

- To estimate the incubation period using this method, click on the 'Incubation Period (Bayesian)' window and download the mock dataset at the bottom of the page.
- For this method, check that the dataset has the minimum and maximum Exposure dates for each entry
- Upload the dataset and choose the distribution (lognormal, Weibull, and gamma) of interest.
- Select the number of iterations (between 500 and 4000) and click on 'Run Bayesian Model' to run the model.

Note: The model uses Bayesian MCMC (rstan). Calculations typically take between 30-45 seconds.



How to choose the model to use for the Maximum likelihood estimation

For an initial analysis, users are recommended to select “ALL” rather than choosing a single distribution at the onset. This allows EpiParam to fit the lognormal, Weibull, and gamma distributions to the same incubation-period data and enables direct comparison of their model-fit statistics and fitted distribution plots. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) can then be used to guide selection of the most appropriate model for final interpretation and reporting.

The AIC and BIC compare the relative fit of candidate models while accounting for model complexity. Lower AIC and BIC values indicate better relative model fit among models fitted to the same dataset. Therefore, after selecting “ALL,” users should compare the AIC and BIC values across the fitted distributions. When both criteria favor the same distribution, and its fitted curve provides a reasonable representation of the observed data, that distribution can generally be selected as the preferred model for reporting.

The recommended workflow is as follows;

1. Select “ALL” for the initial analysis to fit all available candidate distributions.
2. Compare the AIC and BIC values for the lognormal, Weibull, and gamma models.
3. Identify the model with the lowest AIC and BIC values as having the strongest relative statistical support among the models compared.
4. Examine the distribution plots to determine whether the fitted curve adequately represents the observed incubation-period data, including the central portion and upper tail of the distribution.

5. Consider whether the estimated distribution and parameters are epidemiologically plausible for the disease under investigation.
6. Select the best-supported model for the final estimation, interpretation, and reporting of the incubation-period parameters.

If the model with the lowest AIC also has the lowest BIC, model selection is relatively straightforward. However, if AIC and BIC favor different distributions, or if their values are very similar across models, users should not select a model based solely on the smallest numerical value. The fitted distribution plots, epidemiological plausibility, study objectives, and findings from previous studies should also be considered.

How to interpret the results

The estimated incubation-period distribution describes the expected time between exposure or infection and symptom onset among the cases included in the analysis. EpiParam provides fitted distribution, model parameters, summary measures, and model-fit statistics to support interpretation.

The mean incubation period represents the average estimated time from exposure to symptom onset according to the fitted model. For example, an estimated mean incubation period of 5.2 days indicates that, on average, symptom onset is estimated to occur approximately 5.2 days after exposure among individuals represented by the fitted distribution.

The variance describes the degree of variability in incubation periods. A larger variance indicates greater variation in the time from exposure to symptom onset, whereas a smaller variance indicates that incubation periods are more concentrated around the estimated mean.

The 95th percentile represents the incubation-period duration below which approximately 95% of incubation periods are expected to fall according to

the fitted model. For example, a 95th percentile of 11.8 days means that the fitted distribution estimates that approximately 95% of incubation periods are 11.8 days or shorter, while approximately 5% are longer. This quantity may be useful when considering the expected upper range of incubation periods, although public-health decisions such as quarantine duration should not be based on this estimate alone.

The distribution plot should be interpreted together with the numerical estimates. The histogram represents the observed incubation periods in the uploaded dataset, while the fitted curve represents the incubation-period distribution estimated by the selected statistical model. Close agreement between the fitted curve and the observed distribution provides visual support for the model fit; substantial discrepancies may indicate that the selected distribution does not adequately characterize the observed data.

When multiple distributions are fitted, differences in the estimated mean, variance, and 95th percentile should be considered alongside the AIC, BIC, and fitted distribution plots. Particular attention should be given to differences in the upper tail because the estimated 95th percentile may be more sensitive to the choice of distribution than the estimated mean.

Results from subgroup analyses should be interpreted as estimates for the selected subgroup only. For example, selecting females aged 20–29 years restricts the analysis to observations meeting both criteria. Estimates from small subgroups may be unstable and should therefore be interpreted cautiously.

3.0 Serial interval (SI)

The serial interval is the time between successive cases in a disease chain of transmission. The serial interval is estimated from the interval between clinical onsets, in which case it is the 'clinical onset serial interval' when these quantities are observable.

Data preparation.

Similar to the incubation period;

- Save your data spreadsheet in a '.csv' format example:
"Busan_data.csv"
- In the csv data spread sheet, the following variables should be present with the given specificities (Capital/small letter sensitive);
 - ✓ *Date of exposure of the disease → rename as "date_of_exposure"
 - ✓ *Date of symptom onset → rename as "date_of_onset"
 - ✓ *Source of infection → rename as "source_of_infection"
 - ✓ Gender/Sex → rename as "sex"
 - ✓ Age/Age-group → rename as "age_group"
 - ✓ Any other variable depending on researchers' interests

Mock data for the COVID-19 outbreak.

Busan_real.csv - LibreOffice Calc

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A	B	C	D	E	F	G	H	I	J	K
1	id	date_of_exposure	regions	date_of_onset	Source of infection	type	country	Transmission type	Age group	sex
2	#1	2021-11-24	incheon	2021-11-25	11.24 Arrivals from Nigeria(ET672)	abroad	Nigeria	Imported	41M	
3	#2	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	45F	
4	#3	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	43M	
5	#4	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	35F	
6	#5	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	51M	
7	#6	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	43F	
8	#7	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	61M	
9	#8	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	33M	
10	#9	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	42F	
11	#10	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40F	
12	#11	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	34F	
13	#12	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	55F	
14	#13	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	44M	
15	#14	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	46M	
16	#15	2021-11-25	Daejeon	2021-11-26	Patient #1	National	Domestic	Domestic	31F	
17	#16	2021-11-26	Daejeon	2021-11-26	Patient #15	National	Domestic	Domestic	47M	
18	#17	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	39F	
19	#18	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	48M	
20	#19	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	51F	
21	#20	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	46M	
22	#21	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	53M	
23	#22	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	48F	
24	#23	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	56F	
25	#24	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	44F	
26	#25	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	21F	
27	#26	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	25M	
28	#27	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	47M	
29	#28	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	48F	
30	#29	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	37M	
31	#30	2021-11-26	Daejeon	2021-11-29	Patient #29	National	Domestic	Domestic	42F	
32	#31	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	43M	
33	#32	2021-11-26	Daejeon	2021-11-30	Patient #1	National	Domestic	Domestic	59F	
34	#33	2021-11-26	Daejeon	2021-11-30	Patient #1	National	Domestic	Domestic	51M	

Required formats

- ❖ date_of_exposure
This is the probable date of exposure of the case, in the format “YYYY-MM-DD”
- ❖ date_of_onset
This is the probable date of symptom onset of the disease, in the format “YYYY-MM-DD”
- ❖ id as id. A numeric identity which is given to each data unit
- ❖ source_of_infection
The most probable source of infection of an individual given rise to the ‘infector – infectee’ pair
- ❖ sex
the gender of the individuals; All = Everyone, regardless of gender; Male = Male; Female = Females

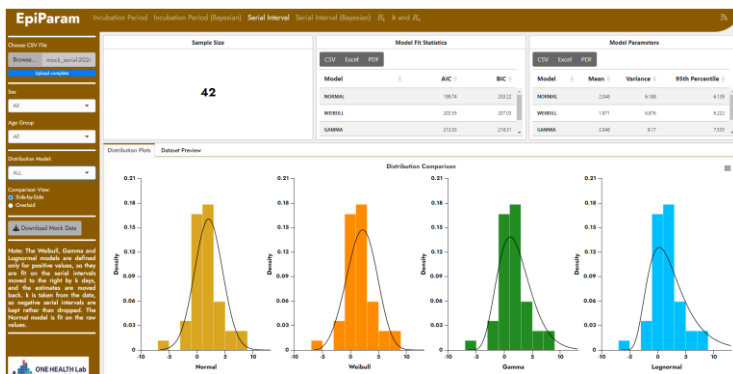
❖ age_group

The ages of the participants in years and should be imputed as a whole number. Classifications of the age into age groups include: All = Everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

A) Estimating the Serial Interval using parametric models

In order to estimate the serial interval;

1. Click on the ‘serial interval’ icon at the top of the App.
2. Upload the .csv file containing the cleaned data you wish to use, by clicking on “Browse” and selecting the data from the computer.
3. This will produce an output as follows;



Output Description (SI)

The output window is made up of 5 sub-windows.

- The ‘Sample Size’ window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.

- The 'Dataset Preview' window previews the uploaded dataset.
- The 'Model Fit Statistics' window that provides the AIC and BIC of each model.
- The 'Model Parameters' window that provides estimates (mean, variance, 25th, 75th, and 95th quantile) for each model.
- The 'Data Plots' window that demonstrates the output histogram of the distribution.

Sub-group analysis

Based on the researcher's needs, select stratifications by sex or by age group, or a combination of both.

B) Estimating the Serial Interval using the Bayesian parametric models

In order to estimate the serial interval using the Bayesian method;

1. Click on the 'Serial Interval (Bayesian)' icon at the top of the App.
2. Upload the .csv file containing the cleaned data you wish to use, by clicking on "Browse" and selecting the data from the computer. The data here should contain the minimum and maximum onset dates of the infectors and infectees.
3. Upload the dataset and choose the distribution (lognormal, Weibull, normal and gamma) of interest.
4. Select the number of iterations (between 500 to 4000) and click on 'Run Bayesian Model' to run the model.

Note: The models estimates the serial interval by linking the infectors to the infectees. Therefore, ensure that 'Source of infection' IDs match the id. Calculations typically take between 30-45 seconds.

How to choose the number of iterations

The number of iterations determines how many sampling steps are used by the Bayesian model to approximate the posterior distributions of the parameters. EpiParam allows users to select between 500 and 4,000 iterations. Increasing the number of iterations generally provides more samples for posterior estimation but also increases computation time.

For an initial analysis, users may begin with a lower or moderate number of iterations (e.g., 1,000–2,000) to explore the model and assess its performance. For the final analysis and reporting, a higher number of iterations (e.g., 4,000) is recommended, particularly when the posterior or trace plots obtained with fewer iterations do not appear stable.

However, the number of iterations alone does not determine whether a Bayesian model has converged. Users should examine the trace plots and posterior distributions produced by EpiParam before interpreting the results.

A recommended workflow is:

1. Begin with approximately 1,000–2,000 iterations for an initial model run.
2. Examine the trace plots for the estimated parameters.
3. If the traces appear stable and well mixed, the model results may be examined further.
4. If the traces show substantial trends, prolonged drifts, or poor mixing, increase the number of iterations and rerun the model.
5. For the final analysis and reporting, use a sufficiently large number of iterations, preferably 4,000 within the current EpiParam settings, and confirm that the resulting trace plots show satisfactory sampling behavior.

6. Increasing the number of iterations should not be used as the sole solution to poor model behavior. Persistent instability may indicate problems with the data, model specification, selected distribution, or other aspects of the analysis.

How to interpret the results

The Bayesian serial-interval module estimates the time between symptom onset in an infector and the corresponding infectee, while accounting for uncertainty in the observed onset dates. Results should be interpreted by considering the estimated serial-interval parameters, their uncertainty, the posterior distributions, and the trace plots together.

The mean serial interval represents the estimated average time between symptom onset in the infector and symptom onset in the infectee. For example, an estimated mean serial interval of 4.5 days indicates that, on average, symptom onset in infectees occurs approximately 4.5 days after symptom onset in their infectors.

The variance describes the variability in serial intervals across transmission pairs. A larger variance indicates greater variation in the timing of symptom onset between infectors and infectees, whereas a smaller variance indicates that the observed intervals are more concentrated around the estimated mean.

The quantiles describe the distribution of estimated serial intervals. For example, the 25th and 75th percentiles indicate the values below which 25% and 75% of serial intervals are expected to occur, respectively. The interval between these two values represents the central 50% of the fitted serial-interval distribution. The 95th percentile represents the value below which approximately 95% of serial intervals are expected to occur.

The posterior distributions show the range of parameter values supported by the observed data and the Bayesian model. The concentration and spread of each posterior distribution provide information about the estimated parameter and its uncertainty. A narrower posterior distribution generally indicates greater precision, whereas a wider distribution indicates greater uncertainty.

The trace plots should be examined to assess the sampling behavior of the Bayesian model. A satisfactory trace plot should fluctuate around a relatively stable region without persistent upward or downward trends or prolonged periods in which the sampler remains confined to a limited range. Persistent trends, poor mixing, or substantial instability may indicate that additional iterations or further investigation of the data and model specification is required.

Negative serial intervals can occur and should not automatically be considered errors. A negative serial interval indicates that symptom onset in the infectee occurred before symptom onset in the identified infector, which may occur when transmission takes place before the infector develops symptoms. The accuracy of the onset dates and infector–infectee linkage should nevertheless be checked before interpreting negative values epidemiologically.

Results from subgroup analyses apply only to the selected population. Estimates based on small numbers of infector–infectee pairs may have greater uncertainty and should be interpreted cautiously.

Finally, serial-interval estimates depend on the quality of the transmission-pair and symptom-onset data.

4.0 The effective reproduction number (R_t)

The R_t represents the average number of secondary infections generated by an infectious individual at a particular time t , under the prevailing population conditions. Unlike the basic reproduction number (R_0), which describes transmission in a fully susceptible population, R_t can change over the course of an outbreak as population susceptibility, behavior, interventions, and other factors change.

In EpiParam, R_t is estimated from time-series incidence data and the serial-interval distribution using the method of Cori et al. The application produces estimates of R_t over time together with 95% credible intervals, allowing users to examine changes in transmission during an outbreak.

Data preparation

Save the incidence dataset in CSV (.csv) format. The dataset should contain the dates of reported cases and the corresponding number of cases observed on each date.

The required variables should follow the variable names and formats specified in the mock dataset provided in EpiParam. Users can download an example dataset by clicking “Download Mock Data” on the R_t page.

Required formats

date

The date corresponding to the reported incidence. Dates should be entered in the format:YYYY-MM-DD

incidence

The number of new cases observed on the corresponding date. Incidence should be entered as a non-negative numeric count.

The incidence time series should be arranged chronologically and checked for missing dates, duplicate dates, or incorrect case counts before analysis.

Estimating the effective reproduction number (R_t)

To estimate R_t :

1. Click the “ R_t and R_0 ” window at the top of the EpiParam application.
2. Under Incidence Data, select “Upload a file.”
3. Click “Browse” and upload the prepared CSV incidence dataset.

4. Specify the serial interval. Users can enter the mean and standard deviation (SD) of the serial interval or use the relevant input option provided by the application.
5. Enter the mean serial interval in days.
6. Enter the standard deviation of the serial interval in days.
7. Select the sliding window, which determines the time window over which R_t is estimated.
8. EpiParam automatically estimates R_t and displays the results in the output windows.

The serial-interval values should preferably be obtained from epidemiological data for the outbreak under investigation or from reliable, published evidence for the same or a closely related disease and epidemiological setting.

Choosing the sliding window

The sliding window specifies the number of consecutive days over which transmission is assumed to remain sufficiently stable for estimating R_t . EpiParam allows the user to adjust this value using the sliding-window control.

A shorter window can provide estimates that respond more rapidly to changes in transmission but may result in greater variability and wider uncertainty, particularly when daily case numbers are small. A longer window generally produces smoother and more stable estimates but may obscure rapid changes in transmission.

The choice of window should therefore consider the study objective, incidence levels, duration of the outbreak, and expected speed of changes in transmission. Users should avoid selecting a window solely because it produces a smoother R_t curve.

When comparing R_t estimates across locations or periods, the same estimation settings should preferably be used unless there is a justified reason to use different settings.

Output description

The R_t output contains summary indicators, graphical results, estimates, and the uploaded data.

Cases: Displays the total number of cases represented in the uploaded incidence dataset.

Serial Interval: Displays the serial-interval value used in the R_t estimation.

Latest R_t : Displays the most recent estimate of the effective reproduction number based on the selected sliding window.

R_t Over Time: Displays the estimated effective reproduction number throughout the study period. The central line represents the mean R_t estimate, while the shaded region represents the 95% credible

interval. The horizontal reference line at $R_t = 1$ assists with interpretation of whether transmission is increasing or decreasing.

Incidence: Displays the epidemic curve based on the uploaded incidence data, allowing users to examine changes in reported cases over time.

Serial Interval Distribution: Displays the serial-interval distribution used in estimating R_t .

Estimates: Provides the numerical R_t estimates and associated uncertainty for the corresponding time periods.

Dataset Preview: Allows users to inspect the incidence dataset used in the analysis.

How to interpret the results

The estimated R_t should be interpreted in relation to the reference value of 1.

- $R_t > 1$: Each infectious individual is generating, on average, more than one secondary infection. This is consistent with increasing transmission.
- $R_t = 1$: Each infectious individual generates approximately one secondary infection, indicating approximately stable transmission.

- $R_t < 1$: Each infectious individual generates, on average, fewer than one secondary infection. If this pattern is sustained, transmission is expected to decline.

For example, an R_t estimate of 0.87 indicates that each infectious individual is estimated to generate, on average, approximately 0.87 secondary infections under the conditions prevailing during that estimation period. However, the 95% credible interval should always be considered together with the point estimate.

If the credible interval lies entirely above 1, the results provide stronger evidence that transmission was increasing during that period. If it lies entirely below 1, the results provide stronger evidence that transmission was declining. If the credible interval includes 1, the available data provide greater uncertainty about whether transmission was increasing or decreasing.

The R_t curve should also be interpreted over time rather than from a single estimate alone. Sustained changes are generally more informative than isolated daily fluctuations. For example, a sustained decline from $R_t > 1$ to $R_t < 1$ may indicate a reduction in transmission during the outbreak.

Interpreting uncertainty

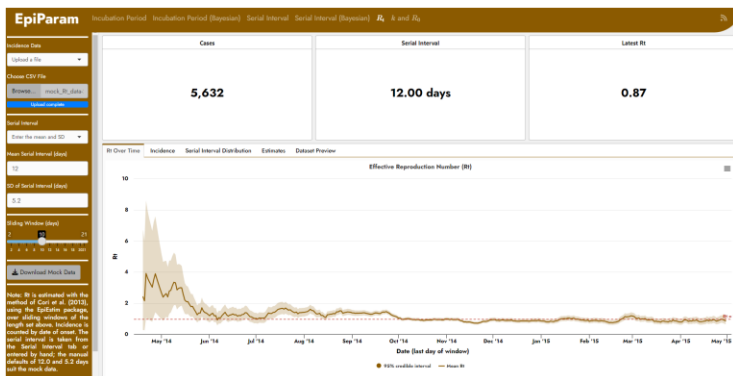
The shaded 95% credible interval represents uncertainty surrounding the estimated R_t . Wider credible intervals indicate greater uncertainty, whereas narrower intervals indicate greater precision.

Uncertainty may be particularly large when case numbers are low, especially near the beginning or end of an epidemic time series. Consequently, very high or low R_t estimates occurring during periods with few cases should be interpreted cautiously.

Important considerations

R_t estimates are influenced by the quality of incidence data, assumptions about the serial interval, and the selected sliding window. Reporting delays, changes in testing or surveillance practices, missing cases, imported cases, case-definition changes, and incomplete reporting can affect the observed incidence curve and consequently the estimated R_t .

The serial-interval assumptions can also influence the magnitude and temporal pattern of R_t . Users should therefore report the mean and SD of the serial interval, the source of these values, and the sliding-window length alongside the R_t estimates.



5.0 Kappa (K) and R_e

In the context of understanding and modelling transmission dynamics, the term "kappa" is sometimes used to represent the degree of variation in infectiousness among infected individuals. This variation in infectiousness is essential to understanding how different individuals may contribute differently to the spread of the disease.

Interpretation

A higher kappa value suggests greater variation, meaning that some individuals are highly infectious (super spreaders), while others have lower infectiousness. A lower kappa value implies less variation, where most individuals have similar levels of infectiousness.

Note: The "kappa" and its associated values can vary depending on the modelling approach. The range of kappa values typically varies from 0 to 1, where:

Kappa = 0 - Implies no variation in infectiousness among individuals. All infected individuals are equally infectious.

Kappa = 1 - Indicates the highest level of variation in infectiousness, where there is extreme heterogeneity among infected individuals.

Data preparation for parameter estimation

- A) Save your data spreadsheet in a '.csv' format
- B) In the CSV data spreadsheet, the following variables should be present in your dataset;
 - ✓ Age - Age of the participants as whole numbers
 - ✓ Gender - Gender of the participants
 - ✓ Date of onset - Date of onset of symptoms for the respective disease
 - ✓ Date of confirm - Date of laboratory confirmation for the respective disease
 - ✓ Source of infection ID - The ID number of the individual from which the infection was gotten.
 - ✓ Cluster_id - Assigned ID of the infection cluster (A,B,C... etc or any other assigned style)
 - ✓ Cluster character - Name or label that is given to the cluster
 - ✓ Infectee generation - the number of successive generation of infections from a primary case in a specific cluster.

Note: A mock dataset has been provided for your information. You can download the mock data in the ‘Download Mock data’ tab in the app.

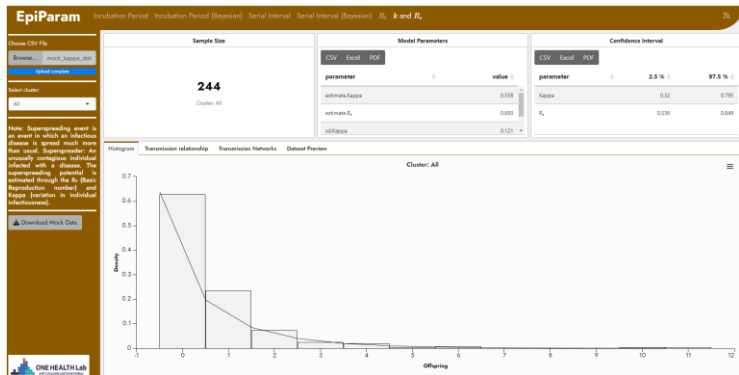
Required formats

- ❖ Date of confirm – Date of laboratory confirmation in the format “DD-MM-YYYY”
- ❖ Date of onset - This is the probable date of symptom onset of the disease, in the format “DD-MM-YYYY”
- ❖ Gender
The gender of the individuals; All = Everyone, regardless of gender; Male = Male; Female = Females
- ❖ Age
The ages of the participants in years and should be imputed as a whole number. Classifications of the age into age groups include; All = Everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

Estimating the Kappa (K) and R_0

To estimate the K and R_0 ;

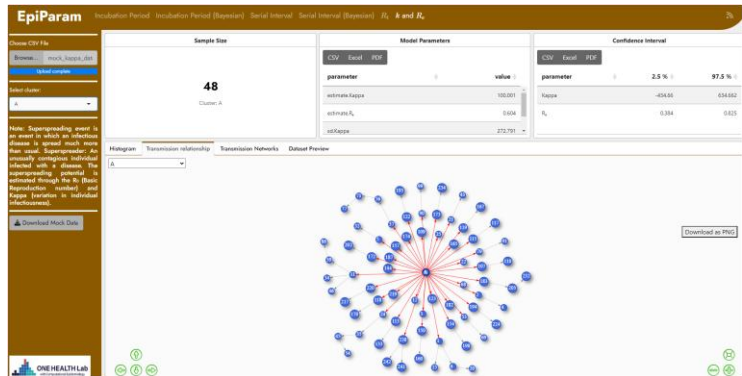
5. Click on the ‘ K and R_0 ’ icon at the top of the App.
6. Upload the .csv file containing the cleaned data you wish to use, by clicking on “Browse” and selecting the data from your device.
7. This will produce an output as follows;



Output Description

The output window is made up of 7 sub-windows.

- The 'Sample Size' window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.
- The 'Dataset preview' window previews the uploaded dataset.
- The 'Model parameter' window that produces estimated K and R_e values as well as the corresponding standard deviations (SD).
- The 'Confidence Interval' window provides the confidence interval of each of the parameters.
- 'Histogram window' demonstrates the output histogram of the distribution.
- The 'Transmission relationship' window demonstrates the transmission relationship of each cluster. This can be viewed by selecting the cluster ID in the dropdown section of the transmission relationship window, as shown below;



- The 'Transmission networks' window demonstrates the network connections among the infector and the infectee.

Sub-group analysis

Based on individual needs, select stratifications by sex or by age group or a combination of both.

Resources

We recommend these resources to support informed use of EpiParam.

Topic	Resource	Description	Link
Outbreak investigation and interpretation	WHO Outbreak Toolkit	Presents outbreak investigation tools and microlearning resources.	https://www.who.int/emergencies/outbreak-toolkit
General applied epidemiology and R	The Epidemiologist R Handbook (R4EPIs)	Covers basics, epidemic modelling, contact tracing, transmission chains, outbreak detection	https://epirhandbook.com/en
General Outbreak Epidemiology	WHO National Outbreak Response Handbook	Provides broader context for how epidemiological analysis fits within outbreak investigation and response.	https://www.who.int/publications/b/73921
EpiNow2	EpiNow2	Useful for understanding where incubation/generation-time distributions come from, including	https://epiforecasts.io/EpiNow2/

		literature estimates versus estimation from individual-level line-list data.	
EpiEstim	EpiEstim	Useful for understanding the serial interval	https://cran.r-project.org/web/packages/EpiEstim/vignettes/demo.html
Bayesian/MCMC convergence	Stan – Posterior Analysis	Explains MCMC, convergence, effective sample size, and related concepts	https://mc-stan.org/docs/reference-manual/analysis.html
R-hat and effective sample size	Stan/RStan convergence diagnostics	Explains why users should assess convergence rather than simply choosing an arbitrary number of iterations.	https://mc-stan.org/rstan/reference/Rhat.html